

UMelCodec: An Ultra-Low-Bitrate and Lightweight Mel-Domain Neural Speech Codec

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Abstract—Ultra-low-bitrate speech coding is crucial for bandwidth-constrained communication and compact storage, but preserving intelligibility, naturalness, and fine-grained acoustic details at bitrates of only a few hundred bps remains challenging. Existing purely acoustic codecs generally operate at relatively high bitrates, whereas incorporating semantic representations can reduce the bitrate, but at the cost of increased complexity. Therefore, we propose UMelCodec, a purely acoustic neural speech codec operating in the mel domain at an ultra-low bitrate of 325 bps with a lightweight architecture. UMelCodec consists of two stages. In the first mel-domain coding stage, alias-free Snake-ConvNeXt (AF-SC) encoder-decoder backbones model temporal and harmonic structures, a factorized single-codebook vector quantizer (VQ) enables ultra-low-rate quantization, and a multi-scale mel-spectrogram discriminator (MSMD) improves mel-spectrogram reconstruction. In the second waveform synthesis stage, a lightweight modified discrete cosine transform (MDCT)-based neural vocoder predicts the MDCT spectrum from the reconstructed mel spectrogram and recovers the waveform through inverse MDCT. Experiments on LibriTTS dataset demonstrate that UMelCodec outperforms strong baselines across most objective reconstruction metrics while maintaining competitive subjective naturalness. In terms of model complexity and inference efficiency, UMelCodec has only 19.67 M parameters, requires 3.85 G MACs, and supports real-time inference on both GPU and CPU platforms.

Index Terms—ultra-low-bitrate speech coding, mel spectrogram, MDCT spectrum, neural vocoder, lightweight

I. INTRODUCTION

SPEECH coding aims to represent speech signals with compact bitstreams and reconstruct intelligible and natural waveforms at decoder side. It is a key technique for bandwidth-constrained communication, speech storage and etc.

Traditional speech codecs rely on carefully designed signal-processing modules. Opus [1] and EVS [2] provide robust speech quality but generally operate at bitrates of several kbps or higher. Recent acoustic-level neural codecs improve rate-distortion performance through learned signal representations. Among them, waveform-domain codecs, e.g., SoundStream [3], EnCodec [4], DAC [5], and BigCodec [6], directly model time-domain signals using encoder-quantizer-decoder architectures, but their deeply stacked downsampling and up-sampling layers incur substantial computational cost. Spectral-domain codecs reduce the modeling burden by exploiting

structured time-frequency representations: APCodec [7], MDCTCodec [8], and Spectral Codec [9] quantize amplitude-phase spectral, modified discrete cosine transform (MDCT) spectral, and mel-spectral representations, respectively. However, these codecs still operate at relatively high bitrates, and preserving fine-grained acoustic details in the few-hundred-bps regime remains challenging. To address this bitrate limitation at the acoustic level, FMelCodec [10] introduces flow-matching-based refinement [11] for sub-kbps mel-spectrogram coding, but its iterative mel refinement and waveform generation require multiple inference steps and increase computational complexity. Alternatively, semantic-assisted codecs, e.g., SemantiCodec [12] and FocalCodec [13], achieve sub-kbps coding by leveraging high-level representations from large pretrained models; however, the additional upstream models substantially increase the overall model size and computational cost and may discard fine-grained acoustic information. Therefore, jointly achieving an ultra-low bitrate, low computational complexity, and high acoustic fidelity remains a challenge.

To address these limitations, we develop **UMelCodec**, an Ultra-low-bitrate and lightweight **Mel**-domain neural speech **Codec** that relies solely on acoustic representations and operates at only 325 bps. UMelCodec employs a two-stage architecture. The mel-domain coding stage uses alias-free Snake-ConvNeXt (AF-SC) encoder-decoder backbones, a factorized single-codebook vector quantizer (VQ), and a multi-scale mel-spectrogram discriminator (MSMD) to compress and reconstruct mel spectrograms at an ultra low bitrate. The waveform synthesis stage then employs a lightweight vocoder to predict MDCT spectra from the reconstructed mel spectrograms and recover the waveforms. Experiments show that UMelCodec surpasses strong baselines on most objective reconstruction metrics while achieving competitive subjective naturalness. Moreover, UMelCodec maintains low model complexity and computational cost while supporting real-time inference across different hardware platforms.

II. PROPOSED METHOD

A. Overview

As illustrated in Fig. 1(a), the proposed UMelCodec is a two-stage ultra-low-bitrate speech coding framework. Let $\mathbf{x} \in \mathbb{R}^T$ and $\hat{\mathbf{x}} \in \mathbb{R}^T$ denote the input and reconstructed waveforms sampled at f_s , respectively, where T is the number of waveform samples. In the first mel-domain coding stage, a mel spectrogram $\mathbf{M} \in \mathbb{R}^{N \times D}$ is first extracted from $\mathbf{x}$ using a hop size of H samples, yielding $N = T/H$ mel frames, where D is the number of mel bins. The mel-domain codec then maps $\mathbf{M}$ to a discrete index sequence $\mathbf{d} = [d_1, d_2, \dots, d_{N'}]^T \in \mathbb{N}^{N'}$

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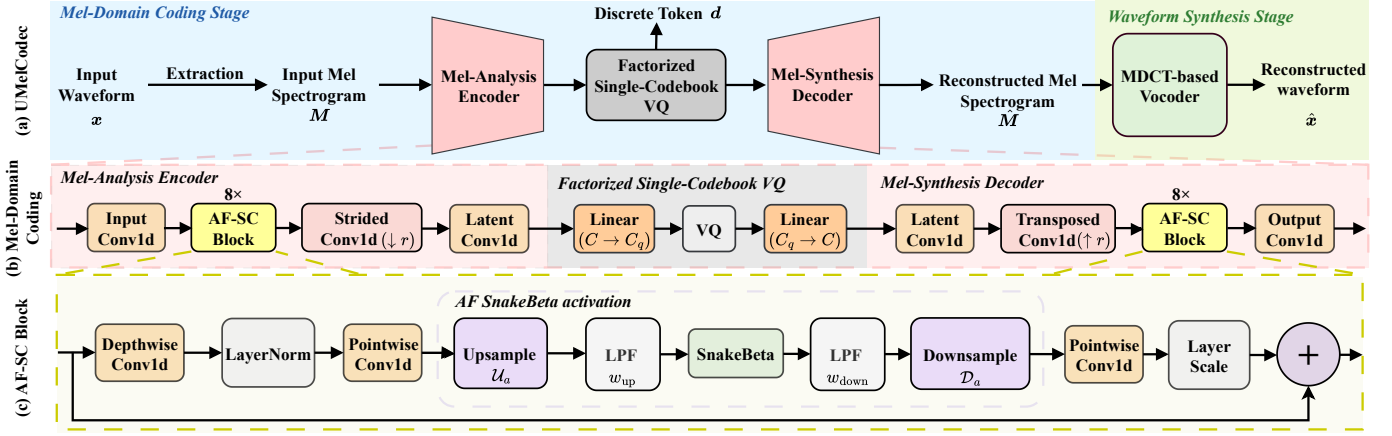


Fig. 1. Overall framework of the proposed UMelCodec.

and reconstructs mel-spectrogram $\hat{M} \in \mathbb{R}^{N' \times D}$, where each $d_n \in \{1, \dots, K\}$, K is the codebook size, and N' is the number of latent frames after downsampling. In the second waveform synthesis stage, an MDCT-based vocoder reconstructs $\hat{x}$ from $\hat{M}$.

B. Mel-Domain Coding Stage

As shown in Fig. 1(b), the mel codec consists of a mel-analysis encoder, a factorized single-codebook VQ, and a mel-synthesis decoder. Given an input mel spectrogram M , the mel-analysis encoder first applies an input convolution to map each D -dimensional mel frame into a higher-dimensional feature space. The resulting sequence is processed by eight AF-SC blocks to model temporal dependencies and periodic acoustic structures. A strided convolution then reduces the temporal frame rate by an overall factor of r , yielding $N' = N/r$ latent frames. Finally, a latent convolution reduces the feature dimension and produces the latent sequence $Z \in \mathbb{R}^{N' \times C}$ for quantization, where C denotes the latent feature dimension.

As shown in Fig. 1(c), AF-SC block follows a ConvNeXt-style architecture [14], [15]. A large-kernel depthwise convolution first models temporal context, followed by layer normalization and a pointwise projection that expands the feature dimension. An alias-free (AF) SnakeBeta activation [16], [17] introduces a periodic inductive bias for modeling harmonic structures, after which a second pointwise projection restores the input dimension. Layer scaling and a residual connection are further applied to stabilize optimization and preserve input information. AF-SnakeBeta applies the SnakeBeta nonlinearity at an increased temporal resolution, with fixed low-pass filters (LPFs) placed before and after the activation:

$$\text{AF-SnakeBeta}(\mathbf{f}) = \mathcal{D}_a(\mathbf{w}_{\text{down}} * \text{SnakeBeta}(\mathbf{w}_{\text{up}} * \mathcal{U}_a(\mathbf{f}))), \quad (1)$$

where $\mathbf{f}$ denotes the input feature sequence, $\mathcal{U}_a(\cdot)$ and $\mathcal{D}_a(\cdot)$ denote upsampling and downsampling by a factor of a , respectively, and $*$ denotes convolution. The fixed Kaiser-windowed sinc LPF kernels $\mathbf{w}_{\text{up}}$ and $\mathbf{w}_{\text{down}}$ suppress the spectral images introduced by upsampling and remove the high-frequency components generated by SnakeBeta before downsampling, respectively, thereby reducing aliasing.

The latent sequence Z produced by the mel-analysis encoder is subsequently discretized using the factorized single-codebook VQ. Following [5], each C -dimensional latent vector is projected to a lower-dimensional quantization space of dimension C_q , where $C_q < C$, and matched to its nearest entry in a codebook of K vectors. The resulting frame-level indices form the discrete sequence d . The selected codebook vectors are then projected back to dimension C before being passed to the mel-synthesis decoder. This $C \rightarrow C_q \rightarrow C$ factorization facilitates code assignment under the single-codebook setting. The resulting theoretical bitrate is $R = \frac{f_s}{Hr} \log_2 K$ (bps).

The mel-synthesis decoder reverses the transformations of the mel-analysis encoder. A latent convolution first expands the quantized representation to the decoder feature dimension. A transposed convolution then upsamples the temporal sequence by a factor of r , restoring the original mel frame rate and sequence length N . The upsampled features are further refined by eight AF-SC blocks, and an output convolution maps them back to D mel bins to obtain the reconstructed mel spectrogram $\hat{M}$.

C. Waveform Synthesis Stage

The neural waveform synthesis stage reconstructs the time-domain waveform from the decoded mel spectrogram using a lightweight MDCT-based vocoder. First, an amplitude-spectrum estimate is derived from $\hat{M}$ using the pseudo-inverse of the mel filter-bank matrix. The estimated spectrum is then processed by an input projection and eight ConvNeXt v2 blocks [15] to model temporal and spectral dependencies, followed by an output projection that predicts the corresponding MDCT spectrum. Finally, the reconstructed waveform $\hat{x}$ is obtained through inverse MDCT.

D. Training Strategy

The modules in mel-domain coding stage and neural waveform synthesis stage are trained separately. For the mel-domain codec, the training objective comprises reconstruction, quantization, and adversarial losses. The mel-spectrogram reconstruction loss $\mathcal{L}_{\text{rec}}$ measures the equal-weight ℓ_1 and ℓ_2 distances between M and $\hat{M}$. Following VQ-VAE [18], the quantization objective includes a commitment loss $\mathcal{L}_{\text{com}}$ and

a codebook loss $\mathcal{L}_{cb}$. To improve the perceptual and structural fidelity of the reconstructed mel spectrograms, we employ an MSMD comprising three sub-discriminators at progressively downsampled temporal resolutions to distinguish real and reconstructed mel spectrograms. The resulting adversarial loss $\mathcal{L}_{adv}$ and feature-matching loss $\mathcal{L}_{fm}$ encourage realistic mel-spectral patterns and stabilize adversarial training, respectively. The overall objective is

$$\mathcal{L}_G = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{com}\mathcal{L}_{com} + \lambda_{cb}\mathcal{L}_{cb} + \lambda_{adv}\mathcal{L}_{adv} + \lambda_{fm}\mathcal{L}_{fm}, \quad (2)$$

where λ_{rec} , λ_{com} , λ_{cb} , λ_{adv} , and λ_{fm} are hyperparameters.

The MDCT-based vocoder is trained using an MDCT-domain multi-scale discriminator, which processes MDCT spectra at three resolutions through parallel two-dimensional convolutional branches. Following the multi-scale adversarial learning strategy [19], [20], its objective combines adversarial and feature-matching losses with MDCT-spectrum and mel-spectrogram reconstruction losses.

III. EXPERIMENTAL SETUPS

A. Dataset

We conducted experiments on the LibriTTS corpus [21], with all speech clips downsampled to $f_s = 16$ kHz. For training, we used the full official training set, including *train-clean-100*, *train-clean-360*, and *train-other-500*. The UMelCodec was evaluated on *test-clean* and *test-other* to assess reconstruction quality under clean and more challenging acoustic conditions.

B. Implementation

For UMelCodec¹, mel spectrograms were extracted from waveforms using an FFT size of 1024, a window length of 640, a hop size of $H = 160$, and $D = 80$ mel bins. For the AF-SC blocks in both the mel-analysis encoder and mel-synthesis decoder, the feature and intermediate dimensions were set to 256 and 512, respectively, while the resampling factor a in the AF-SnakeBeta activation was set to 2. The temporal downsampling factor was set to $r = 4$. The latent dimension, quantization dimension, and codebook size were set to $C = 32$, $C_q = 8$, and $K = 8192$, respectively, corresponding to a theoretical bitrate of 325 bps. The loss weights λ_{rec} , λ_{com} , λ_{cb} , λ_{adv} , and λ_{fm} were set to 45, 2.5, 10, 0.1, and 0.1, respectively. AdamW [22] was used with a learning rate of 2×10^{-4} , $\beta_1 = 0.8$, $\beta_2 = 0.99$, and an exponential decay factor of 0.999. The separately trained MDCT-based vocoder contained eight ConvNeXt v2 blocks, with feature and intermediate dimensions of 513 and 1536, respectively. The MDCT spectra were computed using a hop size of 160 and 320 frequency bins.

C. Baselines

We compare UMelCodec with DAC [5], BigCodec [6], FMelCodec [10], SemantiCodec [12], and FocalCodec [13]. Since DAC, BigCodec, and FMelCodec do not provide official

325-bps checkpoints, we retrain them on the same dataset using a 25-Hz frame rate and a single 8192-entry codebook. For SemantiCodec and FocalCodec, we use their official 325-bps checkpoints.

D. Evaluation Metrics

We evaluated speech reconstruction quality using both objective and subjective metrics. The objective metrics include the Virtual Speech Quality Objective Listener (ViSQOL) [23], log-spectral distance (LSD), mel-cepstral distortion (MCD) [24], and short-time objective intelligibility (STOI) [25]. For subjective evaluation, mean opinion score (MOS) tests were conducted on Amazon Mechanical Turk, where each batch contained 20 utterances rated by at least 25 native English listeners on a five-point naturalness scale with 0.5-point increments. Model complexity and inference efficiency were evaluated on the LibriTTS test-other set using the number of parameters (Param.), multiply-accumulate operations (MACs), and real-time factor (RTF). MACs were calculated using a 2-second speech input, while GPU and CPU RTFs were measured on an NVIDIA GeForce RTX 3090 and an Intel Xeon E5-2680 v3 processor, respectively.

IV. EXPERIMENTAL RESULTS

A. Comparison with Baseline Codecs

Table I compares UMelCodec with representative acoustic codecs, including waveform- and mel-domain systems, as well as semantic codecs, all operating at 325 bps. UMelCodec achieves the best ViSQOL, MCD, and STOI results on both test sets, together with the best LSD on test-clean and the joint-best LSD on test-other. Compared with the waveform-domain DAC and BigCodec, UMelCodec provides higher perceptual quality, intelligibility, and spectral fidelity, demonstrating the effectiveness of directly coding compact mel representations. It also outperforms the generative acoustic codec FMelCodec without requiring iterative flow-matching refinement. As a purely acoustic codec, UMelCodec achieves higher full-reference reconstruction fidelity than SemantiCodec and FocalCodec without relying on large pretrained representation models, suggesting that semantic representations may preserve linguistic content at the expense of some fine-grained speaker and acoustic information.

The subjective results further support the objective findings to some extent. On test-clean, UMelCodec achieves subjective naturalness comparable to FocalCodec, with no statistically significant difference according to a t -test with $p = 0.32$. On test-other, FocalCodec obtains a higher MOS despite its lower full-reference objective performance. This may be because its pretrained SSL representations suppress part of the background interference and generate cleaner speech rather than strictly preserving the original acoustic environment. In contrast, UMelCodec maintains competitive subjective naturalness while more faithfully preserving the input acoustic characteristics.

In terms of model complexity and inference efficiency, UMelCodec requires only 19.67 M parameters and 3.85 G MACs, both the lowest among the evaluated systems. According to the RTF measurements, it achieves inference speeds of

¹Audio samples can be found at <https://github.com/redmist328/UMelCodecdemo>.

TABLE I

OBJECTIVE AND SUBJECTIVE EVALUATION RESULTS OF UMElCODEC AND BASELINE CODECS AT 325 BPS ON THE LIBRITTS TEST-CLEAN AND TEST-OTHER SETS. GPU AND CPU RTFS ARE REPORTED, WHERE $\alpha\times$ DENOTES AN INFERENCE SPEED OF α TIMES REAL TIME. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. MOS VALUES ARE REPORTED WITH 95% CONFIDENCE INTERVALS.

Model	test-clean					test-other								
	ViSQOL ↑	LSD (dB)↓	MCD (dB)↓	STOI ↑	MOS ↑	ViSQOL ↑	LSD (dB)↓	MCD (dB)↓	STOI ↑	MOS ↑	Param. (M)↓	MACs (G)↓	GPU RTF ↓	CPU RTF ↓
DAC	3.17	1.04	4.10	0.79	3.40 (±0.08)	2.91	1.19	4.67	0.75	2.86 (±0.08)	76.71	95.59	0.022 (45.58×)	1.546 (0.65×)
BigCodec	3.38	1.00	3.47	0.83	3.53 (±0.07)	3.14	1.16	3.96	0.80	3.21 (±0.08)	114.76	89.24	0.030 (32.88×)	4.311 (0.23×)
FMelCodec	3.56	1.02	4.11	0.81	3.33 (±0.08)	3.29	1.06	4.42	0.77	2.51 (±0.08)	30.62	37.51	0.033 (30.75×)	0.713 (1.40×)
SemantiCodec	3.35	1.18	4.20	0.78	3.10 (±0.08)	3.27	1.14	4.39	0.75	2.79 (±0.08)	1877.67	3207.42	1.685 (0.59×)	35.940 (0.03×)
FocalCodec	3.48	1.18	3.70	0.83	3.63 (±0.08)	3.25	1.35	4.55	0.80	3.64 (±0.07)	144.22	17.02	0.011 (91.16×)	0.355 (2.82×)
UMelCodec	3.80	0.96	3.13	0.86	3.59 (±0.08)	3.56	1.06	3.85	0.81	3.30 (±0.08)	19.67	3.85	0.006 (156.37×)	0.120 (8.36×)

TABLE II

ABLATION RESULTS OF UMElCODEC ON THE LIBRITTS TEST-CLEAN SET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY.

Model	ViSQOL ↑	LSD (dB)↓	MCD (dB)↓	STOI ↑
UMelCodec w/o MSMD	3.75	1.00	3.33	0.86
UMelCodec w/o AF filtering	<u>3.76</u>	<u>0.97</u>	<u>3.16</u>	0.85
UMelCodec w/o VQ factorization	3.68	0.99	3.49	0.84
UMelCodec	3.80	0.96	3.13	0.86

156.37 times real time on GPU and 8.36 times real time on CPU, demonstrating efficient deployment on both platforms.

B. Ablation Studies

We conduct ablation experiments to investigate the contributions of the main components in UMElCodec, as reported in Table II. UMElCodec w/o MSMD exhibits lower ViSQOL and higher LSD and MCD than the full model, while achieving the same STOI. This result suggests that multi-scale adversarial supervision mainly improves the perceptual quality and spectral fidelity of the reconstructed mel spectrograms by encouraging realistic temporal structures at different resolutions. UMElCodec w/o AF filtering, which removes the upsampling, downsampling, and low-pass filtering operations while retaining the SnakeBeta activation, performs worse than the full model across all four objective metrics. This demonstrates that suppressing aliasing around the periodic nonlinearity benefits perceptual quality, spectral reconstruction, and speech intelligibility. UMElCodec w/o VQ factorization also exhibits consistent degradation across all metrics, particularly in ViSQOL, MCD, and STOI, indicating that low-dimensional code assignment facilitates effective quantization and preserves perceptually important information under the single-codebook setting.

C. Analysis of the Effect of Codebook Size

The codebook size determines both the representation capacity and the utilization efficiency of the factorized single-codebook VQ. We therefore analyze its effect to justify the choice of 8192 entries and examine the trade-off between reconstruction quality and codebook utilization. As shown in Fig. 2, ViSQOL generally improves as the codebook size increases and reaches its maximum at 8192, where all codebook entries are utilized. Further increasing the codebook

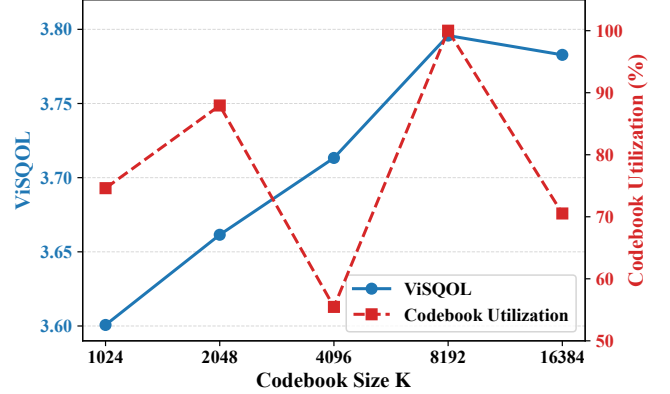


Fig. 2. Effect of the codebook size K on ViSQOL and codebook utilization on the LibriTTS test-clean set.

size to 16384 reduces the utilization to approximately 70% and slightly degrades ViSQOL. In this case, although the nominal codebook size is doubled, the number of utilized entries increases by only about 40% compared with the fully utilized 8192-entry codebook, indicating that many additional entries are redundant or rarely selected. Therefore, enlarging the codebook beyond 8192 provides limited effective capacity while making quantization less efficient and degrading reconstruction quality. Based on this trade-off, a codebook size of 8192 is adopted in UMElCodec.

V. CONCLUSION

This paper presented UMElCodec, an ultra-low-bitrate and lightweight mel-domain neural speech codec operating at 325 bps without external pretrained models or iterative generation. UMElCodec employs AF-SC encoder-decoder backbones, a factorized single-codebook VQ, and an MSMD for mel-spectrogram compression and reconstruction. A lightweight MDCT-based vocoder predicts the MDCT spectrum from mel spectrograms and recovers the waveform through inverse MDCT. Experiments on LibriTTS show that UMElCodec outperforms strong baselines in objective reconstruction quality while maintaining competitive subjective naturalness. The complete codec requires only 19.67 M parameters and 3.85 G MACs and supports real-time GPU and CPU inference. Ablation experiments validate the MSMD, alias-free activation, and VQ factorization, while the codebook-size analysis identifies 8192 entries as a favorable trade-off between reconstruction quality and codebook utilization.

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